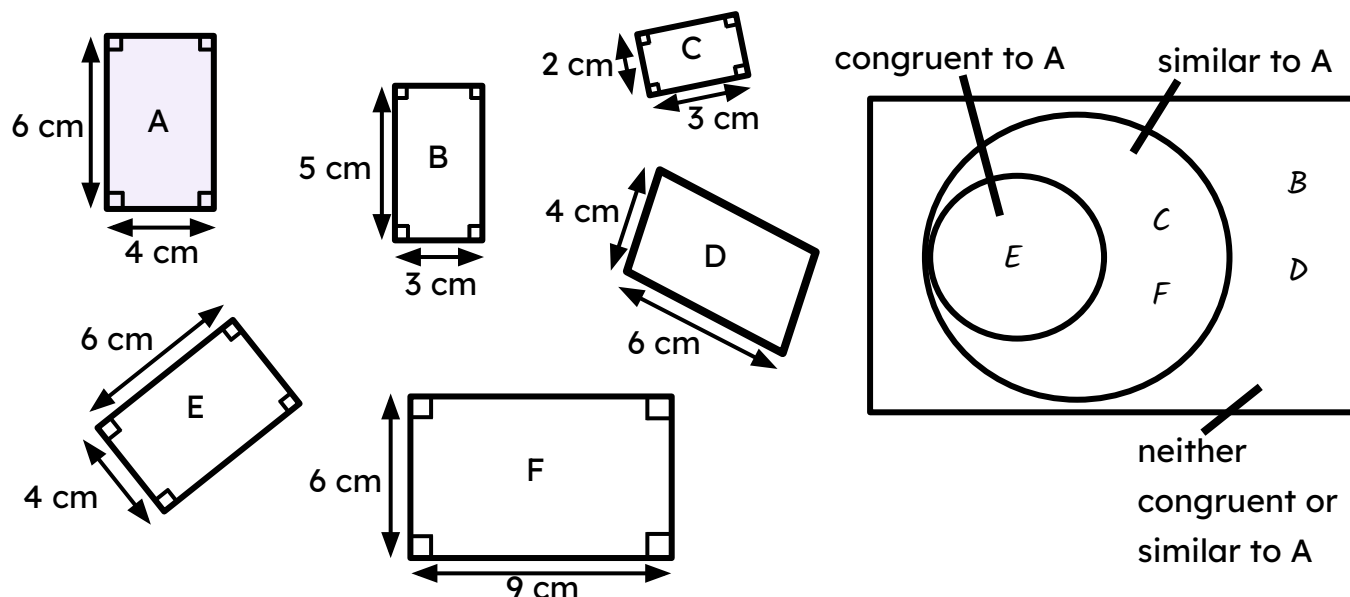




Congruent or similar or neither

Task A: Distinguishing between similarity and congruence

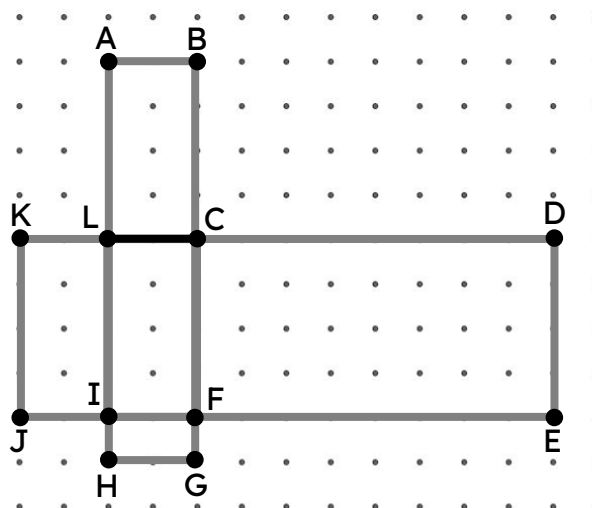
1) Compare each shape with Shape A and write the letters B to F in the Venn diagram.



2) The information describes five connected rectangles, with vertices at points A to L.

- LC is a horizontal line with length 2 units.
- Point F is below Point C and $CF = 2LC$.
- LCFI is a rectangle.
- KLIJ is congruent to LCFI.
- ABCL is congruent to LCFI.
- CDEF is similar to LCFI but not congruent.
- FGHI is similar to LCFI but not congruent.

Use the information to plot points A to L on the grid and draw the five rectangles.



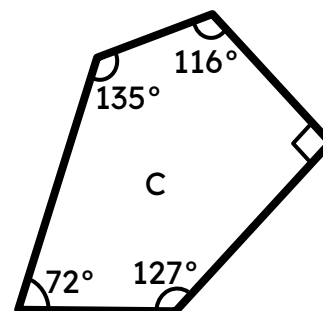
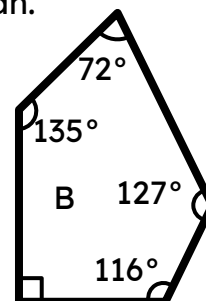
Other solutions are possible if the rectangles overlap, DE are fixed due to space on the grid.



Task B: Checking for similarity and congruence

1) a) Fill in the table for shapes B and C, matching up the angles if you can.

shape A angles	116°	127°	72°	135°	90°
shape B angles	116°	127°	72°	135°	90°
shape C angles	90°	127°	72°	135°	116°



b) Which shape is neither similar nor congruent to Shape A?

shape C

c) Is it possible to determine congruence for the other shape? Explain your answer.

No. Congruence cannot be determined without establishing whether the lengths are the same.

2) a) Fill in the table for shape B.

angles	42°	109°	119°	90°
shape A lengths (m)	6	4	2	8
shape B lengths (m)	12	8	4	16

b) Are shapes A and B congruent, similar or neither?
Justify your answer.

Shape B is similar to shape A. They have the same angles and the lengths in shape B are all double the lengths of shape A.

Shape B is an enlargement of shape A with a scale factor of 2.

